

Effect of Mexiletine on Muscle Stiffness in Patients With Nondystrophic Myotonia Evaluated Using Aggregated N-of-1 Trials

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IMPORTANCE In rare diseases it is difficult to achieve high-quality evidence of treatment efficacy because of small cohorts and clinical heterogeneity. With emerging treatments for rare diseases, innovative trial designs are needed.

OBJECTIVE To investigate the effectiveness of mexiletine in nondystrophic myotonia using an aggregated N-of-1 trials design and compare results between this innovative design and a previously conducted RCT.

DESIGN, SETTING, AND PARTICIPANTS A series of aggregated, double-blind, randomized, placebo-controlled N-of-1 trials, performed in a single academic referral center. Thirty Dutch adult patients with genetically confirmed nondystrophic myotonia (38 patients screened) were enrolled between February 2014 and June 2015. Follow-up was completed in September 2016.

INTERVENTIONS Mexiletine (600 mg daily) vs placebo during multiple treatment periods of 4 weeks.

MAIN OUTCOMES AND MEASURES Reduction in daily-reported muscle stiffness on a scale of 1 to 9, with higher scores indicating more impairment. A Bayesian hierarchical model aggregated individual N-of-1 trial data to determine the posterior probability of reaching a clinically meaningful effect of a greater than 0.75-point difference.

RESULTS Among 30 enrolled patients (mean age, 43.4 [SD, 15.24] years; 22% men; 19 *CLCN1* and 11 *SCN4A* genotype), 27 completed the study and 3 dropped out (1 because of a serious adverse event). In 24 of the 27 completers, a clinically meaningful treatment effect was found. In the Bayesian hierarchical model, mexiletine resulted in a 100% posterior probability of reaching a clinically meaningful reduction in self-reported muscle stiffness for the nondystrophic myotonia group overall and the *CLCN1* genotype subgroup and 93% posterior probability for the *SCN4A* genotype subgroup. In the total nondystrophic myotonia group, the median muscle stiffness score was 6.08 (interquartile range, 4.71-6.80) at baseline and was 2.50 (95% credible interval [CrI], 1.77-3.24) during the mexiletine period and 5.56 (95% CrI, 4.73-6.39) during the placebo period; difference in symptom score reduction, 3.06 (95% CrI, 1.96-4.15; n = 27) favoring mexiletine. The most common adverse event was gastrointestinal discomfort (21 mexiletine [70%], 1 placebo [3%]). One serious adverse event occurred (1 mexiletine [3%]; allergic skin reaction). Using frequentist reanalysis, mexiletine compared with placebo resulted in a mean reduction in daily-reported muscle stiffness of 3.12 (95% CI, 2.46-3.78), consistent with the previous RCT treatment effect of 2.69 (95% CI, 2.12-3.26).

CONCLUSIONS AND RELEVANCE In a series of N-of-1 trials of mexiletine vs placebo in patients with nondystrophic myotonia, there was a reduction in mean daily-reported muscle stiffness that was consistent with the treatment effect in a previous randomized clinical trial. These findings support the efficacy of mexiletine for treatment of nondystrophic myotonia as well as the feasibility of N-of-1 trials for assessing interventions in some chronic rare diseases.

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Public-private partnerships and the interest of industry in smaller niche markets have resulted in an increase of treatments for patients with rare diseases.¹ Small populations and substantial heterogeneity pose considerable challenges to the production of robust evidence of the safety and efficacy of these new treatments in clinical trials.² This challenge has prompted innovative trial designs.³ One of the designs that has been proposed for studying efficacy of drugs for patients with rare diseases is the single-patient, or N-of-1, trial.⁴ Although aggregation of multiple N-of-1 trials holds the potential of efficiently producing evidence of efficacy of treatments,⁵ this approach has not been explicitly compared with a randomized clinical trial (RCT).

Nondystrophic myotonia (NDM) is a rare chronic disease caused by mutations in the skeletal muscle sodium ion (*SCN4A*) or chloride ion (*CLCN1*) channel gene. The key symptom is myotonia, a delayed relaxation of the muscle after voluntary contraction resulting in muscle stiffness. Patients with NDM also experience functionally limiting pain, tiredness, and weakness.⁶

In 2012, an international multicenter crossover RCT showed the clinical effectiveness of mexiletine (a sodium channel blocker) as antimyotonic treatment for patients with NDM in aggregate.⁷ However, in contrast to N-of-1 trials, the RCT was limited by the large within-patient heterogeneity in NDM and did not allow personalized selection of treatment. Both NDM and mexiletine are well suited for an N-of-1 trial design: NDM is a stable chronic disease with symptoms that can be objectively measured, whereas mexiletine is a symptomatic treatment that acts rapidly and quickly subsides on discontinuation.

This study reports results of a series of aggregated placebo-controlled N-of-1 trials of mexiletine in patients with NDM, at both the individual and the group level, including a comparison with results from the previously conducted RCT.

Methods

Clinical Trial Design and Patients

The trial was approved by the regional medical ethics committee, and written informed consent was obtained from all patients. The trial was performed at the Radboud University Medical Center in Nijmegen, the Netherlands. The full trial protocol is available in [Supplement 1](#) and also has been published.⁸ The Radboud Clinical Trial Center was responsible for management of data quality and monitoring of the trial. We performed a series of double-blind, randomized, placebo-controlled N-of-1 trials in adult patients with a clinical phenotype and genetically confirmed diagnosis of NDM, without cardiac or psychiatric comorbidity or comedication (eTable 1 in [Supplement 2](#) contains additional details of the inclusion and exclusion criteria), selected from the Dutch neuromuscular database.⁹ Each N-of-1 trial consisted of 1 to 4 treatment sets, comprising 11 weeks each: a 4-week period of mexiletine and a 4-week period of placebo treatment, block-randomized, with a 1-week washout in between and 2 weeks for statistical interim analysis at the end (eFigure 1 in

Key Points

Question Does a study of mexiletine in patients with nondystrophic myotonia using an aggregated N-of-1 trials design produce efficacy results consistent with those from a randomized clinical trial?

Findings In a series of N-of-1 trials of mexiletine vs placebo in 27 patients with nondystrophic myotonia, there was a reduction in mean daily-reported muscle stiffness of 3.12 of a possible 9 points. This compares with a reduction of 2.69 points in a published randomized clinical trial.

Meaning These findings support the efficacy of mexiletine and also support the potential value of N-of-1 trials for assessing interventions in some chronic rare diseases.

[Supplement 2](#)). The eligibility criteria, intervention protocol, outcome measurements, and follow up scheme were identical to those used in a published, randomized, multicenter, placebo-controlled crossover trial.⁷

Randomization and Masking

Patients were randomly assigned to receive mexiletine hydrochloride (200 mg) capsules, or placebo capsules, 3 times daily. Computer-based randomization was performed by an independent statistician. Randomization was by block for each of the treatment sets (range, 1-4 sets). The Radboud University Medical Center Department of Pharmacy was responsible for the preparation of study medication and the packaging and labeling of treatment kits.

Procedures

Patients receiving antimyotonic treatment underwent a 2-week washout period before baseline. Baseline measurements included sex and age. Patients had 4 to 16 study visits, depending on the number of treatment sets necessary to draw conclusions regarding the treatment effect exceeding the clinically meaningful difference, with a probability greater than 0.80. During each visit, questionnaires, clinical action myotonia bedside tests, handgrip dynamometry, electrophysiological tests, and electrocardiograms were conducted. Venous blood was collected for measurement of mexiletine serum levels.

Adverse events were ascertained by active surveillance during trial visits and passive surveillance. Determination of the relationship between an adverse event and mexiletine treatment was performed by a data and safety monitoring board together with the trial pharmacologist (B.J.S.) and investigator (B.C.S.), who remained blinded to the assigned treatment during the process.

Outcomes

The primary outcome measure was the mean daily self-reported stiffness severity score reported with an interactive voice response (IVR) diary.⁷ The IVR is an automated, centralized telephone response system that records severity and frequency of stiffness (as well as pain, weakness, and tiredness) and that has been validated in patients with myotonia.^{7,10}

Patients noted if they experienced symptoms during the previous 24 hours and rated the severity of the symptoms on an ordinal scale (1-9, with 9 being the worst ever experienced).

Secondary outcomes included mean daily self-reported (using the IVR) severity scores for pain, weakness, and tiredness; the Individual Neuromuscular Quality of Life questionnaire composite score (0-100 scale; a higher score indicates greater disease severity)^{11,12} and 36-Item Short-Form Health Survey (Dutch version) mental and physical component scores (both 0-100 scales; lower score indicates greater disease severity)¹³; the first, fifth, and mean of 5 attempts of myotonic bedside tests: eyelid closure and handgrip muscle relaxation times after forceful muscle contraction for 5 seconds^{7,14}; and the Timed Up&Go test, which measures the time in which the patient rises from a chair, walks 3 meters, turns around, walks back, and sits down again, at a self-selected speed.^{15,16}

For all relaxation time measurements and the Timed Up&Go test, relaxation or test time generally increases with increasing myotonia and higher scores indicate greater disease severity. Handgrip dynamometry measured maximum voluntary isometric contractions (MVICs) of the hand grip and the subsequent 90% to 5% relaxation time (duration between 90% and 5% of muscle relaxation).¹⁷ Additionally, from each MVIC, we analyzed the peak force (0-600 N; a lower score indicates greater disease severity) and the maximal percent decline of this peak force (within the 3-second MVIC measurement) as measure of transient paresis (0%-100%; a higher percentage score indicates greater disease severity). Myotonic discharge grading was based on the presence of myotonic discharges with concentric needle electromyography in the left rectus femoris muscle at rest (10 insertions, with 30 seconds of evaluation per insertion) according to criteria of Streib et al (0-3, with 0 indicating no myotonic discharges and 3 indicating myotonic discharges with every insertion; a higher score indicates greater disease severity).¹⁸

All secondary outcomes (apart from daily IVR measurements) were assessed at both the beginning and end of each 4-week treatment set, with the exception of myotonic discharge grading measurements that were only assessed at the end of each 4-week treatment set. After each treatment set, the patients' preference for 1 of the 2 treatment periods within the treatment set was noted. All random serum mexiletine blood levels were analyzed at the end of the trial.¹⁹

Statistical Analysis

Bayesian Analysis

Based on clinical experience (consensus meeting with 3 clinical experts), a 0.75-point difference was considered a clinically meaningful difference for all 4 IVR scores. This would correspond to a 20% change and an effect size of 0.43 based on a previous natural history study that showed a mean of 3.85 (SD, 1.75) on the IVR stiffness scale in untreated patients with NDM.^{10,20}

On completion of each treatment set, a Bayesian analysis was conducted to calculate the posterior probability of mexiletine producing a clinically meaningful difference in the individual patient (eMethods 1 in Supplement 2). The patient and the treating physician were advised to discontinue the N-of-1

trial if the posterior probability of a meaningful clinical treatment effect was higher than 80% (discontinue trial participation and start regular treatment) or lower than 20% (discontinue trial participation and do not start regular treatment). In all other cases, advice was given to continue the N-of-1 trial.

To combine the results of the multiple N-of-1 trials, a Bayesian hierarchical model was used (eMethods 2 and 3 in Supplement 2).⁵ The N-of-1 trial results from each patient were aggregated into a sample mean and variance, assuming a normal distribution centered around the patient's true mean effect and variance. At the next levels of aggregation (analyses of prespecified genotype subgroup and total NDM patients group), patients' mean effect sizes were modeled, assuming a normal distribution around the genotype subgroups (eMethods 3) or the overall NDM group mean (eMethods 2), with between-patient variance. In the Bayesian hierarchical model, the estimates for each genotype subgroup were informed by information from all patients, in light of their genotype. All patients who completed at least 1 treatment period were included within the aggregated analysis, with inclusion of all available data. Analysis was performed using Winbugs version 1.4 (BUGS Project [<https://www.mrc-bsu.cam.ac.uk/software/bugs/>]), run from R statistics version 3.1.3 (R Project for Statistical Computing).

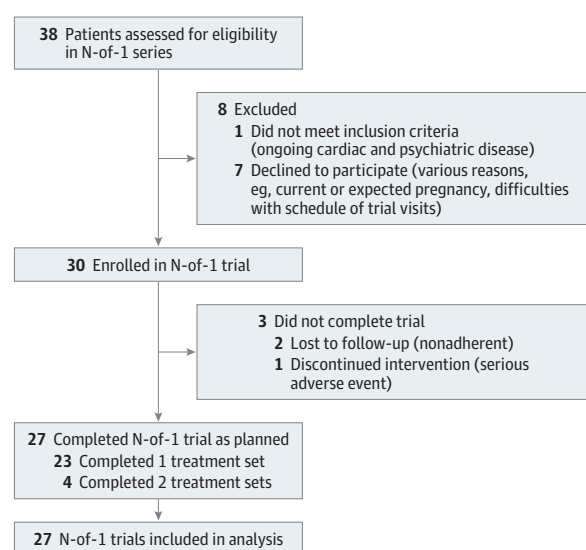
To estimate simulation-based type I error and bias attributable to an observed-response-based stopping rule, a post hoc simulation-based sample size calculation (with virtual data from 1000 aggregated N-of-1 trials) was performed (eMethods 4 in Supplement 2). This simulation revealed that a sample size of 30 patients, with an estimated mexiletine treatment effect of 1.75 (based on preliminary results of the previous RCT,⁷ provided by the study team), resulted in a simulation-based power of 69% with a bias in estimation observed through simulation of -0.46 (ie, the mean treatment effect of 1.29 derived from simulated data sets minus the assumed fixed treatment effect of 1.75).

Frequentist Analysis

To compare results from our trial with those from the previous RCT, we performed frequentist analysis of our N-of-1 trials data with an approach similar to that used in the previous RCT. A linear mixed-effects model was used for the primary outcome measure (IVR stiffness score) and secondary outcome measures (IVR tiredness, pain, and weakness scores), with the following candidate covariables: treatment, genotype, mean baseline IVR score, randomization order, and period effect (in patients who underwent multiple treatment sets), with genotype $\times$ treatment interaction as fixed effect. Variance components were used as covariance structure, selecting covariables with $P < .10$. This resulted in exclusion of randomization order and period effect in the final model. Daily-reported IVR severity scores were replaced with treatment-period means from the last 2 weeks in a treatment period, and the assumed normal distribution was confirmed.

For all other secondary outcome measures that were normally distributed, we used dependent t tests to calculate mean treatment effects, significance levels, and confidence intervals for the total NDM group level. Additionally, we tested for

Figure 1. Recruitment and Patient Flow



genotype $\times$ treatment interaction for each secondary outcome and report significant differences in treatment effects among genotype subgroups.

All P values were 2-sided, and $P < .05$ was considered statistically significant for all tests. Since a primary outcome measure was used, all other P values presented were for secondary outcome measurements and were not adjusted for multiple testing. Therefore, secondary outcome measures should be interpreted as exploratory. Adverse events are reported as counts and percentages. Analyses were performed using SPSS version 22 (IBM/SPSS), with inclusion of all available data.

Comparison of Results With Previous RCT

To study the comparison between the treatment effect of our aggregated N-of-1 trials and that of the previously reported RCT,⁷ we used the aggregated N-of-1 trial treatment effect from the frequentist analysis (instead of the Bayesian analysis) to ensure maximal compatibility and avoid introduction of additional differences.

Results

Recruitment and Patient Flow

Eligible patients were recruited between January 6, 2014, and February 2, 2015 (Figure 1). The first patient was enrolled on February 21, 2014, and the last study visit took place on June 10, 2015. Of the 38 patients contacted and recruited, 1 was ineligible (because of ongoing cardiac and psychiatric disease) and 7 declined participation for a variety of reasons (current or expected pregnancy, difficulties with schedule of trial visits). Thirty patients were randomized and received study medication. There were 3 dropouts: 2 patients did not complete study visits, and for 1 patient the individual N-of-1 trial was stopped because of a serious adverse reaction (Figure 1).

Baseline Data

Twenty-two men and 8 women with a mean age of 43.4 years (SD, 15.24; range, 19–65 years) were enrolled. Nineteen patients had a mutation in the skeletal muscle chloride channel gene (*CLCN1*) and 11 patients had a mutation in the skeletal muscle sodium channel gene (*SCN4A*) (eTable 2 in Supplement 2). IVR stiffness scores (higher in patients with *CLCN1* genotype), IVR pain scores (higher in patients with *SCN4A* genotype), and eyelid closure action myotonia scores (higher in patients with *SCN4A* genotype) differed between the 2 genotype subgroups at baseline (Table 1).

Numbers Analyzed

Of the 27 patients who completed their individual N-of-1 trial, 23 underwent a single treatment set and 4 completed a second treatment set; thus, in total, 31 treatment sets from 27 patients were analyzed. For the outcome assessments, 773 of 868 (89%) telephone calls to assess the primary outcome were completed and 2676 of 2728 (98%) possible outcome measures for the secondary outcomes were collected at the in-person visits. Since the amount of missing data was relatively small and assumed missing at random, multiple imputation was not performed.

Primary Outcome Measure

In 24 of the 27 patients (89%), Bayesian analysis of the individual N-of-1 trial data showed the predefined clinically meaningful effectiveness of mexiletine. In these patients, the individual N-of-1 trial was stopped and mexiletine treatment was continued in a normal clinical care setting. In 3 patients (11%), Bayesian analysis showed the predefined clinical ineffectiveness of mexiletine. Their individual N-of-1 trials were stopped and mexiletine treatment was discontinued. All 3 mexiletine nonresponders had a *SCN4A* genotype (Figure 2; eTable 3 in Supplement 2).

Bayesian-aggregated N-of-1 trials analysis showed a 100% posterior probability of reaching a clinically meaningful difference for the NDM group overall and for the *CLCN1* genotype subgroup; this probability was 93% for the *SCN4A* genotype subgroup (Figure 3A,B). In the total nondystrophic myotonia group, the median muscle stiffness score was 6.08 (interquartile range, 4.71–6.80) at baseline and was 2.50 (95% credible interval [CrI], 1.77–3.24) during the mexiletine period and 5.56 (95% CrI, 4.73–6.39) during the placebo period. This corresponded with a mean reduction of IVR stiffness score of 3.06 (95% CrI, 1.96 to 4.15) for the NDM group ($n = 27$), 3.84 (95% CrI, 2.52 to 5.16) for the *CLCN1* genotype subgroup ($n = 16$), and 1.94 (95% CrI, 0.35 to 3.53) for the *SCN4A* genotype subgroup ($n = 11$) (Figure 2 and Figure 3A). The claim that mexiletine reduces myotonia with a meaningful difference (with >95% probability) was already reached after aggregating results from the first 11 consecutive patients with NDM (Figure 3C). No significant randomization order effect ($P = .85$) or period effect ($P = .22$) were found.

Comparison of Results With Previous RCT

Using frequentist reanalysis, mexiletine compared with placebo resulted in a mean reduction in daily-reported muscle

Table 1. Baseline Patient Characteristics by Genotype Subgroups

Characteristic	<i>CLCN1</i> (N = 16)	<i>SCN4A</i> (N = 11)	P Value ^a
Age, mean (range), y	50 (24-65)	38 (19-64)	.09
Men, No. (%)	13 (81)	7 (64)	.39
Pretrial antimyotonic medication, No. (%)			
Mexiletine	2 (13)	0	.50
Other	7 (44)	5 (46)	.62
IVR score, median (IQR) ^b			
Stiffness	6.7 (6.0-7.0)	4.7 (3.3-5.1)	.002
Pain	0 (0-1.1)	3.4 (2.3-5.5)	.002
Weakness	3.8 (1.1-5.0)	0.5 (0-6.1)	.06
Tiredness	3.4 (1.4-6.0)	5.1 (3.9-6.1)	.16
Handgrip action myotonia, median (IQR), s ^{c,d}	1.1 (0.9-2.0)	1.5 (1.0-6.4)	.40
Eyelid closure action myotonia, median (IQR), s ^{c,d}	1.2 (0.6-2.9)	4.0 (1.3-11.2)	.01
Timed Up&Go, median (IQR), s ^{c,e}	9.7 (8.8-11.0)	9.5 (8.3-10.3)	.49
INQoL composite score, median (IQR) ^f	84.0 (74.5-110.3)	98.0 (56.0-120.0)	.82
SF-36 score, median (IQR) ^g			
Physical component	40.8 (37.7-46.6)	38.7 (33.6-40.0)	.11
Mental component	52.1 (44.7-57.1)	55.0 (37.2-59.1)	.88

Abbreviations: *CLCN1*, skeletal muscle chloride channel gene; INQoL, Individual Neuromuscular Quality of Life Questionnaire; IQR, interquartile range; IVR, interactive voice response diary; *SCN4A*, skeletal muscle sodium channel gene; SF-36, Short-Form 36-Item Health Status Survey.

^a Nonparametric test of differences between genotype disease subgroups using Mann-Whitney U test.

^b Baseline IVR scores derived from a 2-week period before start of the trial. IVR scores represent severity of daily-reported symptoms (stiffness, pain, weakness, and tiredness) on an ordinal scale (1-9, with 9 being the worst ever experienced).

^c Represents the median (IQR) across all patients, with each patient contributing his or her median score on 5 trials of the measure.

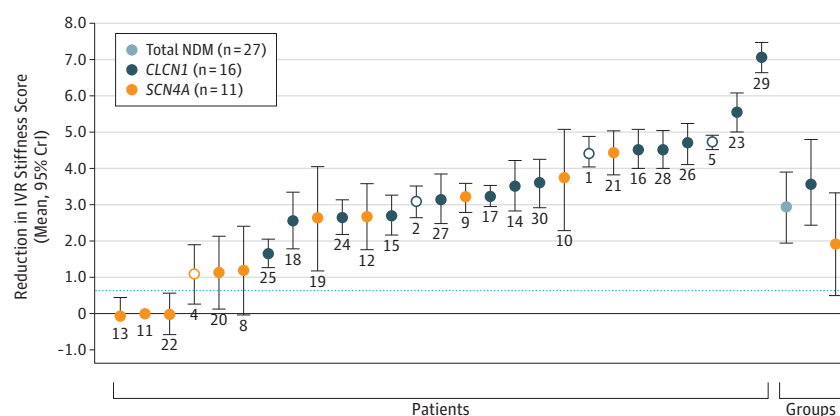
^d Handgrip and eyelid closure action myotonia represent muscle relaxation times after forceful muscle contraction for 5 seconds (relaxation time increases with increasing myotonia).

^e Timed Up&Go test measures the time in which the patient rises from a chair, walks 3 meters, turns around, walks back, and sits down again in a self-selected speed (test time increases with increasing myotonia).

^f Scale, 0 to 100; a higher score indicates greater disease severity.

^g Scale for each component, 0 to 100 scales; a lower score indicates greater disease severity.

Figure 2. Results From Bayesian Analysis at the Individual and Group Level



Primary outcome (interactive voice response [IVR]-reported stiffness) mean treatment effect and 95% credible interval (CrI) for each patient. The estimates for each genotype subgroup (*CLCN1* [skeletal muscle chloride channel gene]; *SCN4A* [skeletal muscle sodium channel gene]) are informed by information from all patients, in light of their genotype. Individual treatment effects (circles) are ordered from smallest to largest effect on the x-axis, with patient numbers

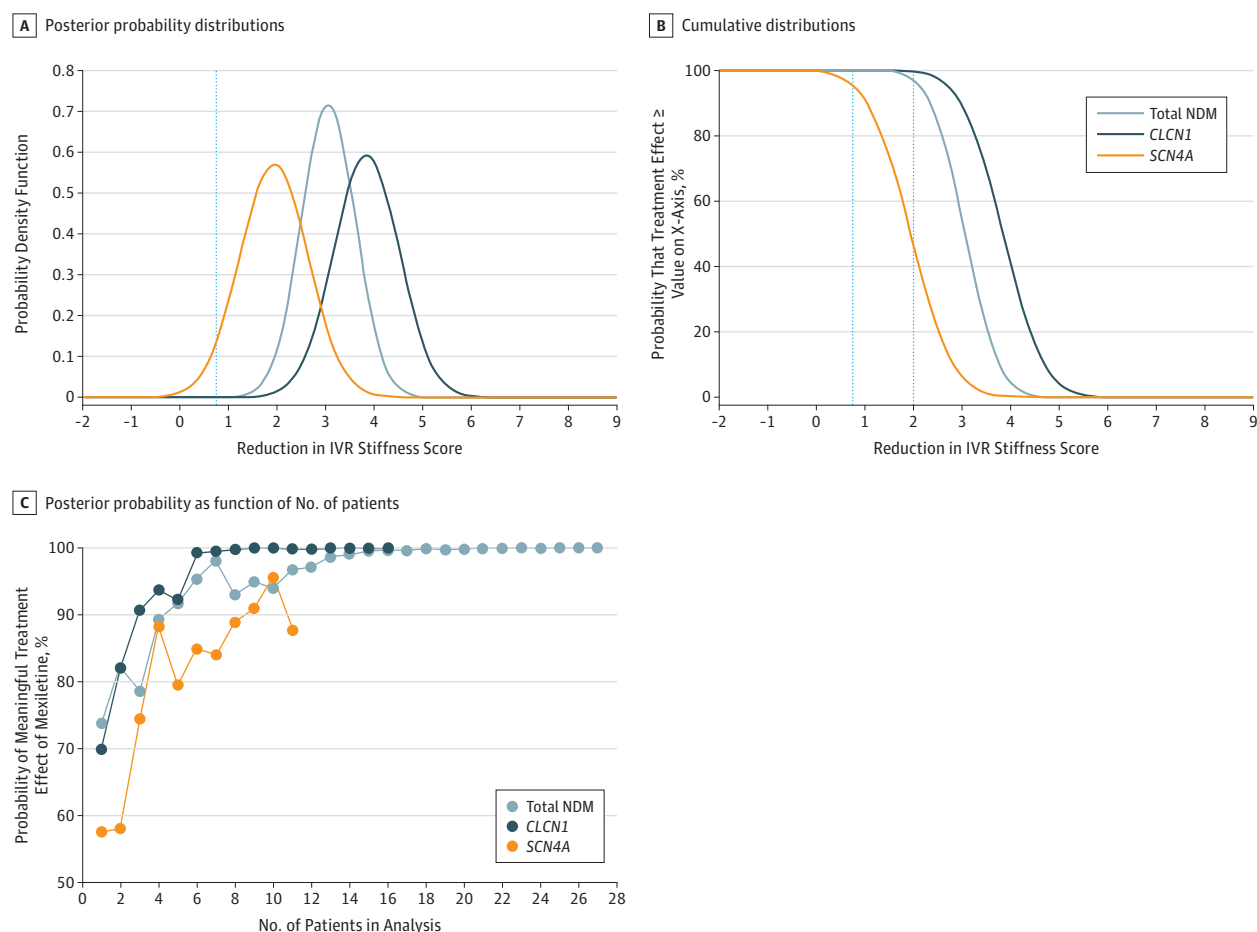
beneath data markers for cross-referencing with demographic and baseline information in eTables 2 and 3 in Supplement 2. Patients 3, 6, and 7 dropped out. Open circles indicate patients who had 2 treatment sets instead of 1 as part of their individual N-of-1 trial (patients 1, 2, 4, and 5). Blue dotted line at $y = 0.75$ represents the threshold for a clinically meaningful treatment effect.

stiffness of 3.12 (95% CI, 2.46 to 3.78) (Table 2), consistent with the previous RCT treatment effect of 2.69 (95% CI, 2.12 to 3.26).⁷

Secondary Outcome Measures

Bayesian-aggregated N-of-1 trials analysis showed a mean treatment effect for IVR-reported pain of 0.68 (95% CrI,

Figure 3. Results From Bayesian Analysis at the Group Level



A, Probability density functions of primary outcome (interactive voice response [IVR]-reported stiffness) treatment effects for the total nondystrophic myotonia (NDM) disease group ($n = 27$) and genotype subgroups (skeletal muscle chloride channel gene [*CLCN1*], $n = 16$; skeletal muscle sodium channel gene [*SCN4A*], $n = 11$). Blue dotted line at $x = 0.75$ represents the threshold for a clinically meaningful treatment effect. Posterior probabilities of a clinically meaningful treatment effect represent the area under the curve to the right of the dotted line. The estimates for each genotype subgroup are informed by information from all patients, in light of their genotype. B, Cumulative representation of panel A into a probability effect function provides readout for the posterior probability belonging to the treatment effect of interest.

As examples, the blue dotted lines provide readouts for posterior probabilities of reaching a clinically meaningful treatment effect of 0.75-point difference (corresponding to a 20% change on the mean baseline IVR stiffness score; effect size, 0.43), and a 2.0-point difference (corresponding to a 52% change on the mean baseline IVR stiffness score; effect size, 1.14) on the IVR scale. C, Posterior probability of reaching a clinically meaningful mexiletine treatment effect as function of the number of consecutive patients used in the Bayesian analysis. Note that a probability of 95% or greater is reached after the first 11 consecutive patients (patients 1-11, as labeled in Figure 2 and in eTables 2 and 3 in Supplement 2) for the total NDM. Numbers on the x-axis do not match those in Figure 2.

−0.52 to 1.89) for the total NDM group ($n = 27$). Mean treatment effects for IVR-reported weakness and tiredness were 1.48 (95% CrI, 0.18 to 2.77) and 1.23 (95% CrI, −0.23 to 2.67). This corresponded with a 45% (IVR-reported pain), 87% (IVR-reported weakness), and 74% (IVR-reported tiredness) probability of reaching a clinically meaningful effect (>0.75-point difference) (eFigure 2 in Supplement 2). Corresponding treatment effects using frequentist analysis were 0.70 (95% CI, 0.18 to 1.23) for IVR-reported pain, 1.56 (95% CI, 1.05 to 2.06) for IVR-reported weakness, and 1.27 (95% CI, 0.58 to 1.95) for IVR-reported tiredness (Table 2).

Results of the additional secondary outcomes based on frequentist analysis are presented in Table 2. Secondary objective clinical and electrophysiological outcome measures

that showed a statistically significant (frequentist) treatment effect at NDM group level included the 36-Item Short Form Health Survey (Dutch version) physical and mental component scores, Individual Neuromuscular Quality of Life questionnaire composite score, mean of handgrip and eyelid closure action myotonia bedside tests, walking speed, handgrip dynamometry peak force, and the myotonic discharges grade on needle electromyography. Significant treatment $\times$ genotype interaction was present (in favor of the *CLCN1* subgroup) for the fifth transient paresis measurement (−23.85 [95% CI, −32.45 to −15.24] for *CLCN1* genotype vs 13.71 [95% CI, −1.96 to 25.47] for *SCN4A* genotype; $P < .001$ for interaction), the mean of 5 transient paresis measurements (−12.37 [95% CI, −18.35 to −6.38] for *CLCN1* genotype

Table 2. Primary and Secondary Outcome Measures for Total Nondystrophic Myotonia Disease Group (N = 27) Using Frequentist (Non-Bayesian) Statistics

Outcome Measure ^a	Baseline Score, Mean (SD) ^b	Placebo Period, Mean (SD)	Mexiletine Period, Mean (SD)	Treatment Effect, Mean (95% CI) ^c	P Value ^d
IVR score (primary outcome) ^e					
Stiffness	5.65 (1.78)	5.55 (2.09)	2.42 (1.81)	3.12 (2.46 to 3.78)	<.001
CLCN1	6.45 (1.71)	6.46 (1.71)	2.60 (1.50)	3.82 (3.10 to 4.54)	<.001
SCN4A	4.34 (2.02)	4.22 (1.93)	2.16 (2.24)	1.89 (1.01 to 2.76)	.002
Genotype × treatment interaction					.004
Pain	1.95 (2.09)	2.08 (2.10)	1.37 (2.13)	0.70 (0.18 to 1.23)	.01
Weakness	2.84 (2.54)	2.96 (2.75)	1.49 (1.66)	1.56 (1.05 to 2.06)	<.001
Tiredness	4.28 (2.28)	3.65 (2.51)	2.41 (2.53)	1.27 (0.58 to 1.95)	.001
Outcome Measure ^a	Baseline Score, Mean (SD) ^b	Change Placebo Period, Mean (95% CI)	Change Mexiletine Period, Mean (95% CI)	Treatment Effect (Placebo-Mexiletine), Mean (95% CI) ^c	P Value ^d
SF-36 ^f					
Physical component score	38.26 (7.81)	1.04 (−0.60 to 2.96)	8.66 (5.94 to 11.38)	7.81 (4.72 to 10.88)	<.001
Mental component score	50.29 (9.67)	−1.85 (−4.81 to 1.11)	4.77 (0.67 to 8.48)	6.78 (1.64 to 11.92)	.001
INQol composite score ^g	96.89 (38.49)	−7.22 (−14.5 to −0.29)	−21.44 (−30.90 to −11.95)	−14.22 (−24.71 to −3.74)	.01
Handgrip action myotonia, s ^h					
First attempt	3.33 (5.00)	0.46 (−0.30 to −1.23)	−2.39 (−4.22 to −0.55)	−2.85 (−5.28 to −0.42)	.02
Fifth attempt	1.36 (1.25)	0.28 (−0.43 to 0.99)	−0.69 (−1.18 to −0.19)	−0.97 (−2.03 to 0.09)	.07
CLCN1	0.93 (0.41)	−0.01 (−0.21 to 0.18)	−0.30 (−0.42 to −0.18)	0.04 (−1.11 to 1.19)	.95
SCN4A	2.59 (3.00)	0.71 (−1.17 to 2.60)	−1.24 (−2.48 to −0.02)	−1.96 (−3.41 to 0.51)	.009
Genotype × treatment interaction					.04
Mean	2.02 (2.33)	0.29 (−0.17 to 0.76)	−1.14 (−1.95 to −0.34)	−1.44 (−2.66 to −0.22)	.02
Eyelid closure action myotonia, s ^h					
First attempt	6.70 (7.38)	−0.74 (−2.49 to 1.01)	−4.99 (−8.25 to −1.73)	−4.25 (−8.45 to −0.05)	.05
Fifth attempt	2.24 (3.23)	−0.44 (−1.24 to 0.36)	−1.39 (−2.57 to −0.21)	−0.95 (−2.25 to 0.35)	.14
Mean	3.50 (4.15)	−0.43 (−1.24 to 0.37)	−2.54 (−4.15 to −0.93)	−2.11 (−3.94 to −0.28)	.03
Timed Up&Go, s ⁱ					
First attempt	10.10 (2.38)	−0.15 (−0.99 to 0.68)	−1.41 (−1.96 to −0.85)	−1.25 (−2.23 to −0.28)	.01
Fifth attempt	8.91 (1.52)	0.29 (−0.45 to 1.03)	−0.70 (−1.13 to −0.27)	−1.00 (−2.01 to 0.03)	.06
Mean	9.51 (1.77)	0.07 (−0.67 to 0.01)	−1.05 (−1.48 to −0.62)	−1.12 (−2.07 to −0.18)	.02
Handgrip dynamometry relaxation time (90%–5%), s ^j					
First attempt	0.84 (1.81)	−0.10 (−0.87 to 0.67)	−0.10 (−0.22 to 0.01)	−0.02 (−0.86 to 0.82)	.96
Fifth attempt	0.31 (0.23)	0.00 (−0.12 to 0.13)	−0.02 (−0.07 to 0.03)	−0.01 (−0.16 to 0.14)	.87
Mean	0.48 (0.70)	−0.07 (−0.33 to 0.19)	−0.05 (−0.10 to 0.00)	−0.02 (−0.28 to 0.25)	.91
Handgrip dynamometry peak force, N ^k					
First attempt	344.07 (173.56)	−7.81 (−39.39 to 23.77)	18.92 (−6.53 to 44.37)	32.92 (−5.10 to 70.94)	.09
Fifth attempt	321.41 (150.51)	6.15 (−14.99 to 27.30)	27.96 (5.27 to 50.65)	24.08 (−5.22 to 53.39)	.10
Mean	336.85 (157.85)	−3.85 (−25.84 to 18.14)	29.90 (5.74 to 54.05)	37.23 (10.19 to 64.28)	.009
Handgrip dynamometry transient paresis, % ^l					
First attempt	29.19 (27.65)	0.69 (−7.68 to 9.07)	−2.4 (−11.55 to 6.75)	−3.7 (−15.10 to 7.68)	.51
Fifth attempt	20.30 (18.75)	0.73 (−3.63 to 5.09)	−8.56 (−17.26 to 0.14)	−12.25 (−22.04 to −2.47)	.02
CLCN1	24.35 (23.11)	4.31 (−1.32 to 9.94)	−18.75 (−28.59 to −8.91)	−23.85 (−32.45 to −15.24)	<.001
SCN4A	9.12 (10.33)	−5.00 (−11.32 to 1.32)	9.56 (1.24 to 17.87)	13.71 (−1.96 to 25.47)	.02
Genotype × treatment interaction					<.001
Mean	24.11 (20.13)	−0.08 (−4.84 to 4.68)	−6.38 (−10.81 to −1.95)	−7.34 (−13.45 to −1.24)	.02
CLCN1	32.56 (22.35)	1.53 (−5.70 to 8.75)	−11.26 (−16.83 to −5.69)	−12.37 (−18.35 to −6.38)	<.001
SCN4A	8.10 (7.40)	−2.64 (−8.50 to 3.22)	2.31 (0.10 to 4.52)	4.46 (−3.85 to 12.76)	.28
Genotype × treatment interaction					.002

(continued)

Table 2. Primary and Secondary Outcome Measures for Total Nondystrophic Myotonia Disease Group (N = 27) Using Frequentist (Non-Bayesian) Statistics (continued)

Outcome Measure ^a	Baseline Score, Mean (SD) ^b	Placebo Period, Mean (SD)	Mexiletine Period, Mean (SD)	Treatment Effect, Mean (95% CI) ^c	P Value ^d
Myotonic discharges on needle EMG (grading) ^m	-	2.52 (0.89)	1.85 (1.13)	0.67 (0.23 to 1.11)	<.001

Abbreviations: *CLCN1*, skeletal muscle chloride channel gene; EMG, electromyography; INQoL, Individual Neuromuscular Quality of Life Questionnaire; IVR, interactive voice response; *SCN4A*, skeletal muscle sodium channel gene; SF-36, Short-Form 36-Item Health Survey.

^a Linear mixed model results are presented for IVR measurements. For other secondary outcome measures, dependent *t* tests were used to calculate mean treatment effects (n = 16 [*CLCN1*], n = 11 [*SCN4A*]).

^b Mean baseline measurement for IVR measurements were calculated from daily calls made in the 14 d of baseline period. For all other measures, baseline measurements were calculated from data from visit 1 (the start of the first treatment period), except for myotonic discharges for which no baseline measurements were available.

^c Outcome measures are also presented on a genotype subgroup level in case of a significant test for interaction between genotype and treatment in the mixed model.

^d All statistically significant findings favored mexiletine treatment.

^e IVR scores represent severity of daily-reported symptoms (stiffness, pain, weakness, and tiredness) on an ordinal scale (1-9; with 9 being the worst ever experienced).

^f Scale for each component, 0 to 100; a lower score indicates greater disease severity.

^g Scale, 0 to 100; a higher score indicates greater disease severity.

^h Handgrip and eyelid closure action myotonia represent muscle relaxation times after forceful muscle contraction for 5 seconds (relaxation time increases with increasing myotonia).

ⁱ Measures the time in which the patient rises from a chair, walks 3 meters, turns around, walks back, and sits down again, at a self-selected speed (test time increases with increasing myotonia).

^j Relaxation times (duration between 90% and 5% of muscle relaxation; test time increases with increasing myotonia), from 5 trials of 3 seconds during maximum voluntary isometric contractions of the handgrip.

^k Peak force (0-600 N, a lower score indicates greater disease severity), from 5 trials of 3 seconds during maximum voluntary isometric contractions of the handgrip.

^l Transient paresis (the maximal decline (%) in peak force as measure; 0 to 100%, a higher score indicates greater disease severity), from 5 trials of 3 seconds during maximum voluntary isometric contractions of the handgrip.

^m Based on the presence of myotonic discharges with concentric needle EMG in the left rectus femoris muscle at rest (10 insertions, with 30 seconds of evaluation per insertion; 0-3 grading, with 0 indicating no myotonic discharges and 3 indicating myotonic discharges with every insertion; a higher score indicates greater disease severity).

vs 4.46 [95% CI, -3.85 to 12.76] for *SCN4A* genotype; *P* = .002 for interaction), and (in favor of the *SCN4A* subgroup) for the fifth handgrip action myotonia measurement (0.04 [95% CI, -1.11 to 1.19] for *CLCN1* genotype vs -1.96 [95% CI, -3.41 to 0.51] for *SCN4A* genotype; *P* = .04 for interaction) (Table 2).

Adverse Events and Mexiletine Drug Levels

The most common adverse event was gastrointestinal discomfort, which occurred in 21 of 30 patients (70%) during mexiletine treatment periods (eTable 4 in Supplement 2). These symptoms were controlled in most patients with lifestyle advice. One serious adverse event—a reversible urticaria-like rash—was determined to be mexiletine-related, and that patient was excluded from the trial (eFigure 3 in Supplement 2).

No clinically relevant electrocardiographic rhythm abnormalities or cardiac conduction interval changes were observed during the course of the trial. The mean mexiletine serum level at the end of mexiletine treatment periods was 1.06 (SD, 0.51) µg/mL (reference antiarrhythmic therapeutic range, 600-1200 mg/d [serum level, 0.5-2.0 µg/mL]). In 2 patients (patients 4 and 13), serum mexiletine levels were in subtherapeutic range (0.11 and 0.27 µg/mL). Mexiletine levels were not detectable at the start of treatment periods (indicating adequate washout periods) and at the end of placebo treatment periods.

Adherence and Blinding

Mean drug adherence was 94%. Drug adherence was less than 75% for only 2 patients (patients 1 and 23). Patients preferred mexiletine treatment over placebo treatment in 26 of the 31 treatment sets (84%).

Follow-up

All 24 mexiletine responders continued mexiletine treatment during the follow-up period that was completed on September 10, 2016 (range, 18-31 months), without adverse events that occasioned discontinuation.

Discussion

In a series of N-of-1 trials of mexiletine vs placebo in patients with NDM, there was a reduction in mean daily-reported muscle stiffness that was consistent with the treatment effect in a previous randomized clinical trial. The aggregated N-of-1 trials with a Bayesian approach offered a number of benefits.²¹ The design allowed for flexibly deciding on the continuation or discontinuation of the N-of-1 trial for individual patients, depending on the nature of the desired evidence and requisite certainty. It also allowed for identification of important individual treatment differences. For example, there appeared to be a difference in the degree of mexiletine response that favored the *CLCN1* genotype. Furthermore, the Bayesian approach to data analysis allowed estimation of the probability that the effect of mexiletine would be clinically meaningful, which may be a more intuitive way of presenting trial outcomes for all end users (patients, physicians, regulatory agencies), compared with the probability that the results are compatible with a null hypothesis in frequentist statistics. In addition, Bayesian probabilistic statistics allowed the calculation of the group-level treatment effect after adding each individual N-of-1 trial. In this trial, the claim that mexiletine reduces myotonia

with a meaningful difference could be made with sufficient certainty after combining the data from the first 11 consecutive patients with NDM. Although other aggregated N-of-1 trials with Bayesian approaches have been conducted,²²⁻²⁶ to the best of our knowledge the study results of this design have never been compared directly with those of an RCT.

On a clinical level, the aggregated N-of-1 trial results suggest that mexiletine reduced muscle stiffness and also reduced tiredness and weakness in patients with NDM. In accordance with the primary outcome measure, most objective and quality-of-life secondary outcome measures also suggested medium to large treatment effect sizes. Although mexiletine is widely used as off-label treatment for patients with various pain syndromes, in this study, the posterior probability of a clinically meaningful difference in the IVR-reported pain outcome measure was lower than 50%. Mexiletine was generally well tolerated, and the treatment-related adverse reaction profile was consistent with previous mexiletine trials.^{7,27,28} Gastrointestinal discomfort was the most frequent reported adverse effect (70%), and no clinically relevant cardiac rhythm or conduction abnormalities were observed.

Limitations

This study has several limitations. First, no minimal clinically important difference (MCID) has been reported for the main outcome measure, so the MCID was determined based on expert opinion.²⁰ Nevertheless, the presentation of the results of the Bayesian analyses allow for an easy readout of

the effectiveness of mexiletine at different MCID thresholds. Second, we found a debinding effect of mexiletine treatment, reflected by the high percentage of patients preferring mexiletine over placebo treatment (84%). This can be attributable to the large clinical effectiveness on the one hand, or the high occurrence of gastrointestinal adverse reactions on the other hand. The interpretation that the changes in the patient-reported primary outcome measures represent real effectiveness of mexiletine is supported by the finding that most secondary outcome measurements (including objective electrophysiological measures) showed significant treatment effects in line with the primary outcome measure. Third, N-of-1 trials may only be feasible for diseases that are chronic or slowly progressive and for testing a treatment with a rapid response, to limit potential period and carry-over effects. For example, a disease-modifying therapy may not be as amenable to this design.

Conclusions

In a series of N-of-1 trials of mexiletine vs placebo in patients with nondystrophic myotonia, there was a reduction in mean daily-reported muscle stiffness that was consistent with the treatment effect in a previous randomized clinical trial. These findings support the efficacy of mexiletine for treatment of nondystrophic myotonia as well as the feasibility of N-of-1 trials for assessing interventions in some chronic rare diseases.

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REFERENCES

- Groft SC. Rare diseases research: expanding collaborative translational research opportunities. *Chest*. 2013;144(1):16-23. doi:10.1378/chest.13-0606
- Haffner ME. Adopting orphan drugs—two dozen years of treating rare diseases. *N Engl J Med*. 2006;354(5):445-447. doi:10.1056/NEJMp058317
- Augustine EF, Adams HR, Mink JW. Clinical trials in rare disease: challenges and opportunities. *J Child Neurol*. 2013;28(9):1142-1150. doi:10.1177/0883073813495959
- Guyatt G, Sackett D, Taylor DW, Chong J, Roberts R, Pugsley S. Determining optimal therapy—randomized trials in individual patients. *N Engl J Med*. 1986;314(14):889-892. doi:10.1056/NEJM198604033141406
- Zucker DR, Schmid CH, McIntosh MW, D'Agostino RB, Selker HP, Lau J. Combining single patient (N-of-1) trials to estimate population treatment effects and to evaluate individual patient responses to treatment. *J Clin Epidemiol*. 1997;50(4):401-410. doi:10.1016/S0895-4356(96)00429-5
- Trivedi JR, Bundy B, Statland J, et al; CINCH Consortium. Non-dystrophic myotonia: prospective study of objective and patient reported outcomes. *Brain*. 2013;136(pt 7):2189-2200. doi:10.1093/brain/awt133
- Statland JM, Bundy BN, Wang Y, et al; Consortium for Clinical Investigation of Neurologic Channelopathies. Mexiletine for symptoms and signs of myotonia in nondystrophic myotonia: a randomized controlled trial. *JAMA*. 2012;308(13):1357-1365. doi:10.1001/jama.2012.12607
- Stunnenberg BC, Woertman W, Raaphorst J, et al. Combined N-of-1 trials to investigate mexiletine in non-dystrophic myotonia using a Bayesian approach: study rationale and protocol. *BMC Neurol*. 2015;15:43. doi:10.1186/s12883-015-0294-4
- van Engelen BG, van Veenendaal H, van Doorn PA, et al. The Dutch neuromuscular database CRAMP (Computer Registry of All Myopathies and Polyneuropathies): development and preliminary data. *Neuromuscul Disord*. 2007;17(1):33-37. doi:10.1016/j.nmd.2006.09.017
- Statland JM, Wang Y, Richesson R, et al; Cinch Consortium. An interactive voice response diary for patients with non-dystrophic myotonia. *Muscle Nerve*. 2011;44(1):30-35. doi:10.1002/mus.22007
- Sansone VA, Ricci C, Montanari M, Apolone G, Rose M, Meola G; INQOL Group. Measuring quality of life impairment in skeletal muscle channelopathies. *Eur J Neurol*. 2012;19(11):1470-1476. doi:10.1111/j.1468-1331.2012.03751.x
- Seesing FM, Drost G, Groenewoud J, van der Wilt GJ, van Engelen BG. Shared medical appointments improve QOL in neuromuscular patients: a randomized controlled trial. *Neurology*. 2014;83(3):240-246. doi:10.1212/WNL.0000000000000588
- Aaronson NK, Muller M, Cohen PD, et al. Translation, validation, and norming of the Dutch language version of the SF-36 Health Survey in community and chronic disease populations. *J Clin Epidemiol*. 1998;51(11):1055-1068. doi:10.1016/S0895-4356(98)00097-3
- Trip J, Drost G, Ginjaar HB, et al. Redefining the clinical phenotypes of non-dystrophic myotonic syndromes. *J Neurol Neurosurg Psychiatry*. 2009;80(6):647-652. doi:10.1136/jnnp.2008.162396
- Podsiadlo D, Richardson S. The timed "Up & Go": a test of basic functional mobility for frail elderly persons. *J Am Geriatr Soc*. 1991;39(2):142-148. doi:10.1111/j.1532-5415.1991.tb01616.x
- Hammarén E, Kjellby-Wendt G, Lindberg C. Quantification of mobility impairment and self-assessment of stiffness in patients with myotonia congenita by the physiotherapist. *Neuromuscul Disord*. 2005;15(9-10):610-617. doi:10.1016/j.nmd.2005.07.002
- Statland JM, Bundy BN, Wang Y, et al; CINCH Consortium. A quantitative measure of handgrip myotonia in non-dystrophic myotonia. *Muscle Nerve*. 2012;46(4):482-489. doi:10.1002/mus.23402
- Streib EW. AAEE minimonograph #27: differential diagnosis of myotonic syndromes. *Muscle Nerve*. 1987;10(7):603-615. doi:10.1002/mus.880100704
- Li S, Liu G, Jia J, et al. Simultaneous determination of ten antiarrhythmic drugs and a metabolite in human plasma by liquid chromatography—tandem mass spectrometry. *J Chromatogr B Analyt Technol Biomed Life Sci*. 2007;847(2):174-181. doi:10.1016/j.jchromb.2006.10.013
- JAMA Guide to Statistics and Methods. JAMA Network website. <https://jamanetwork.com/collections/44042/guide-to-statistics-and-medicine>. Accessed October 29, 2018.
- Spiegelhalter DJ, Abrams KR, Myles JP. *Bayesian Approaches to Clinical Trials and Health-Care Evaluation*. Chichester, United Kingdom: John Wiley & Sons; 2004.
- Huber AM, Tomlinson GA, Koren G, Feldman BM. Amitriptyline to relieve pain in juvenile idiopathic arthritis: a pilot study using Bayesian metaanalysis of multiple N-of-1 clinical trials. *J Rheumatol*. 2007;34(5):1125-1132.
- Mitchell GK, Hardy JR, Nikles CJ, et al. The effect of methylphenidate on fatigue in advanced cancer: an aggregated N-of-1 trial. *J Pain Symptom Manage*. 2015;50(3):289-296. doi:10.1016/j.jpainsymman.2015.03.009
- Nathan PC, Tomlinson G, Dupuis LL, et al. A pilot study of ondansetron plus metopimazine vs. ondansetron monotherapy in children receiving highly emetogenic chemotherapy: a Bayesian randomized serial N-of-1 trials design. *Support Care Cancer*. 2006;14(3):268-276. doi:10.1007/s00520-005-0875-7
- Nikles J, Mitchell G, McKinlay L, et al. A series of N-of-1 trials of stimulants in brain injured children. *NeuroRehabilitation*. 2017;40(1):11-21. doi:10.3233/NRE-161386
- Sung L, Tomlinson GA, Greenberg ML, et al. Serial controlled N-of-1 trials of topical vitamin E as prophylaxis for chemotherapy-induced oral mucositis in paediatric patients. *Eur J Cancer*. 2007;43(8):1269-1275. doi:10.1016/j.ejca.2007.02.001
- Logigian EL, Martens WB, Moxley RT IV, et al. Mexiletine is an effective antimyotonia treatment in myotonic dystrophy type 1. *Neurology*. 2010;74(18):1441-1448. doi:10.1212/WNL.0b013e3181dc1a3a
- Suetterlin KJ, Bugiardini E, Kaski JP, et al. Long-term safety and efficacy of mexiletine for patients with skeletal muscle channelopathies. *JAMA Neurol*. 2015;72(12):1531-1533. doi:10.1001/jamaneurol.2015.2338